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Phase diagram of the two-dimensional Coulomb gas: A thermodynamic scaling Monte Carlo study

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In this work, we report results for the phase diagram of the two-dimensional Coulomb system. We use a recently proposed simulation method, thermodynamic scaling Monte Carlo, to determine the Kosterlitz–Thouless insulator/conductor transition line and the first-order vapor–liquid coexistence curve. The Kosterlitz–Thouless line terminates at the first-order coexistence curve close to the gas–liquid critical point. Our estimate of the vapor–liquid critical point is, $T^* \approx 0.056$, $\rho^* \approx 0.21$, in only modest agreement with previous theoretical estimates and semiquantitative numerical results.

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I. INTRODUCTION

Despite numerous theoretical^{1–7} and numerical^{8–11} investigations, there are several unresolved questions related to the location and character of phase transitions of the two-dimensional classical system with logarithmically interacting charges. This model, referred to as the “Coulomb gas” (CG), is relevant to understanding vortex unbinding in superconducting and superfluid films. It consists of charged hard disks interacting via a logarithmic pair potential. Kosterlitz and Thouless,¹ showed that at low densities the Coulomb gas undergoes a phase transition between a dielectric phase and a conducting phase. The dielectric phase consists of bound pairs and is characterized by a correlation decay similar to that of a dipolar fluid. In contrast, the conducting phase consists of a mixture of pairs and isolated ions and is characterized by finite screening length and exponential correlation decay. As the transition temperature is approached from below, the response of the system to an external applied electric field jumps discontinuously from a finite value to zero. In addition to the Kosterlitz–Thouless transition, there is evidence^{6,7,9,10} that there is a vapor–liquid transition at finite densities and low temperatures. Despite serious theoretical and numerical efforts in the past, the coexistence curve, the location and the character of the resulting critical point are not precisely known.

In this work we use a recently proposed Monte Carlo method, thermodynamic scaling Monte Carlo^{12–14} to obtain the different phases of this model. We explicitly calculate the dielectric response of the low density vapor phase in order to estimate the Kosterlitz–Thouless insulator/conductor transition points. We then show that at low reduced temperatures the Kosterlitz–Thouless transition transforms to first order dielectric vapor/conducting liquid phase coexistence. Our estimate for the critical point is, $T_c^* = 0.056$, $\rho_c^* = 0.21$. The small system size used in this study and the size effects associated with the approach to the critical point, do not allow us to investigate its character. The Kosterlitz–Thouless line emanates from $T^* = 1/4$ at zero density and terminates on the

first order coexistence curve close to the vapor–liquid critical point.

II. MODEL AND PREVIOUS WORK

The model we have considered is a two-dimensional system of N hard particles (disks) of diameter σ , half of them carrying a positive charge $+q$ and the other half $-q$. The disks move in a structureless background of dielectric constant D . The overall density of the system is $\rho = N/V$, where V is the two-dimensional volume (area). The pair interaction potential between particles i and j , separated by distance r_{ij} is defined by

$$u_{ij} = \begin{cases} -\frac{q_i q_j}{D} \ln\left(\frac{r_{ij}}{\sigma}\right) & r_{ij} > \sigma \\ \infty & r_{ij} \leq \sigma \end{cases} \quad (1)$$

The logarithmic interaction is the solution to Poisson’s equation for two dimensions. The model can be described by two nondimensional quantities, reduced density, $\rho^* = \rho \sigma^2$, and reduced temperature, $T^* = k_B T D / q^2$, where k_B is Boltzmann’s constant. For point ions ($\sigma = 0$) a scaling argument,¹⁵ leads to the exact equation of state

$$\frac{p}{\rho k_B T} = 1 - \frac{T_0}{T} \quad (2)$$

with $T_0^* = 1/4$. The same result can also be obtained by standard Debye–Hückel theory in two dimensions. A hard core term is necessary to prevent the system from collapsing at low temperatures.

Since the pioneering work of Kosterlitz and Thouless,^{1,2} there have been many attempts to determine the phase behavior of the CG at finite densities. Caillol and Levesque,⁹ performed Monte Carlo simulations of a CG system on the surface of a sphere and concluded that the expected Kosterlitz–Thouless (KT) transition becomes first order via a tricritical point at moderate densities, but they could not obtain accurate estimates of the phase envelope. First-order behavior was also found by Minnhagen and Wallin⁶ by renormalization group theory. Their results showed that the KT line terminates on the vapor branch of the coexistence curve at a critical end point. Similar behavior was also found in the

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Monte Carlo work of Lee and Teitel¹⁰ for the lattice analogue of the Coulomb gas. Recent theoretical work by Levin *et al.*⁷ based on extensions of Debye–Hückel theory,¹⁶ suggests that the phase diagram of the two-dimensional CG extends to significantly higher reduced temperatures than its three-dimensional equivalent, the restricted primitive model. In Ref. 7, the KT conductor-to-insulator transition line was seen to merge with the first-order phase envelope at a tricritical point located fairly close to the KT transition temperature at zero density. To overcome the slow equilibration problems that the Coulomb force imposes on the system, Manificat and Caillol¹¹ considered a somewhat different version of the Coulomb gas. Their pair potential model consists of a Coulombic term and a Lennard-Jones term with tunable attractive interactions. The phase envelopes obtained in Ref. 11 were largely due to the presence of the Lennard-Jones term and the pure Coulomb gas situation was never reached. In conclusion, despite numerous previous studies of this model, the position of the phase lines, especially of the first-order liquid–vapor coexistence curve are not precisely known. In the present study, we address this gap using a recently proposed Monte Carlo technique.

III. SIMULATION METHODOLOGY

Simulation studies of phase transitions in strongly interacting ionic systems are hampered by the difficulty of exploring configuration space in systems for which the ionic pairs are bound together with energies many times the thermal excitation energy. For two-dimensional systems, intrinsically low interfacial energies make direct simulations of a two-phase system,¹⁷ or simulations in the Gibbs ensemble¹⁸ impractical. Here, we use a new technique, thermodynamic scaling Monte Carlo (TSMC)^{12–14} to obtain the thermodynamic properties of the two-dimensional CG. This method is an extension of a wide class of Monte Carlo techniques known as umbrella sampling methods.¹⁹ The Metropolis Monte Carlo method,²⁰ generates configurations representative of a thermodynamic state at a fixed temperature and density. In contrast, TSMC samples simultaneously a family of thermodynamic states corresponding to different densities and temperatures. Approximately uniform sampling of relevant parts of the phase space is obtained by an appropriate choice of an (unphysical) probability distribution. The main advantage of the method over the previous umbrella sampling techniques is that it provides the free energy as a function of density for several isotherms which is essential for numerical study of phase transitions. In addition, sampling from an unphysical distribution overcomes quasiergodicity traps occurring at the vicinity of phase transitions. For the two-dimensional CG, formation of long chains of alternating charges, similar to those seen in three dimensions for dipolar hard and soft sphere systems²¹ poses significant ergodicity problems to conventional algorithms, while TSMC allows significant changes in the system configuration by sampling different densities and temperatures.

Here, we briefly review the basics of the method. Consider a range of NVT states starting from a reference state at

density $\rho_0 = N/V_0$ and temperature T_0 . It is trivial to show that the excess free energy difference ΔA_{EX} , between any state within the range and the reference state is given by

$$\beta A_{\text{EX}}(N, V, T) - \beta_0 A_{\text{EX}}(N, V_0, T_0) = \Delta(\beta A_{\text{EX}}) = -\ln \frac{\int_1 d\mathbf{r}^N \exp[-\beta U(\mathbf{r}^N; V)]}{\int_1 d\mathbf{r}^N \exp[-\beta_0 U(\mathbf{r}^N; V_0)]}, \quad (3)$$

where $\beta = 1/k_B T$ and the integration is performed over a unit (hyper) volume. In the above integrals, $\mathbf{r}^N$ is the N -particle configuration in scaled coordinates with respect to the box length, $V^{1/d}$, and $U(\mathbf{r}^N; V)$ is the total potential energy. The objective of this new method is the accurate calculation of ΔA_{EX} over the range of interest in a single Monte Carlo simulation. Evaluation of Eq. (3) by Metropolis Monte Carlo sampling at NV_0T_0 will only be reliable in the immediate neighborhood of the reference. However, instead of sampling the canonical distribution, one can imagine a broader probability distribution $\pi = \pi(\mathbf{r}^N)$ that would uniformly generate configurations associated with a family of macroscopic states. It is then straightforward to show that

$$\Delta(\beta A_{\text{EX}}) = -\ln \frac{\langle \exp[-\beta U(\mathbf{r}^N; V)] / \pi(\mathbf{r}^N) \rangle_\pi}{\langle \exp[-\beta_0 U(\mathbf{r}^N; V_0)] / \pi(\mathbf{r}^N) \rangle_\pi}, \quad (4)$$

where $\langle \rangle_\pi$ denotes ensemble average over the π distribution. The problem then reduces to finding a distribution capable of generating N -particle configurations $\mathbf{r}^N$, that would be relevant to a wide range of NVT conditions. Since we want to simultaneously sample several thermodynamic states it is convenient to write¹³

$$\pi(\mathbf{r}^N) = \sum_j w_j \exp[-\beta_j U(\mathbf{r}^N; V_j)]. \quad (5)$$

Each Boltzmann factor will generate configurations associated with its macroscopic state and clearly a linear combination of the different contributions will cover the thermodynamic space of interest. Careful choice of the coefficients w_j will result in uniform sampling over the range of interest. It can be shown^{13,14} that w_j is related to the excess free energy

$$w_j = \exp[+\beta_j A_{\text{EX}}(N, V_j, T_j)]. \quad (6)$$

However, A_{EX} is exactly the quantity we seek. Equations (4), (5), and (6) imply a general approach in order to determine a suitable sampling distribution. Guess a set of weights w_j , perform a run that samples from Eq. (5), determine excess free energies from Eq. (4), get new weights from Eq. (6), and use them in Eq. (5) to improve the π distribution. This iterative procedure can be done in short trial runs. Once a suitable distribution has been obtained, we perform a long run that covers the density and temperature range of interest, and thus accurate free energies can be determined. In addition, we can also obtain the usual canonical ensemble averages. For instance, one can easily show that the canonical ensemble average of the energy of a thermodynamic state is given by

$$\langle U(\mathbf{r}^N; V) \rangle_{NVT} = \frac{\langle U(\mathbf{r}^N; V) \exp[-\beta U(\mathbf{r}^N; V)] / \pi(\mathbf{r}^N) \rangle_\pi}{\langle \exp[-\beta U(\mathbf{r}^N; V)] / \pi(\mathbf{r}^N) \rangle_\pi}. \quad (7)$$

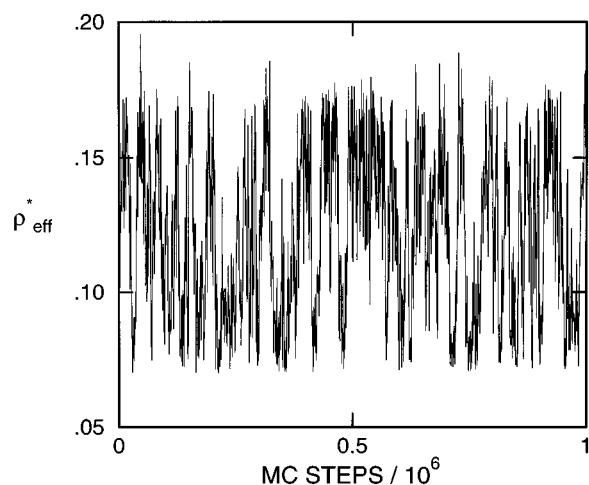


FIG. 1. Evolution of the nearest pair distance expressed as reduced density, for a run covering a range of temperatures from 0.25 to 0.20 and a range of densities from 0.07 to 0.17. $N=50$.

From the practical point of view, we need a quantity that would indicate how uniformly the π distribution covers the phase space of interest. Our system has hard core repulsions and strong attractions that result in many pairs close to contact. A given scaled configuration $\mathbf{r}^N$, contributes most to properties at densities for which it has pairs close to contact at a given (fixed) density. At higher densities there are overlaps and the Boltzmann factor would be zero. We can define an “effective density,” ρ_{eff}^* , based on the minimum pair distance $r_{\min}$ of a configuration,

$$\rho_{\text{eff}}^* = N(r_{\min})^d, \quad (8)$$

where $d=2$ in our specific system. Once again, distances are scaled with respect to the box length $V^{1/d}$. The effective density is a good measure of how adequately we sample configuration space. A case of good, close to uniform sampling is shown in Fig. 1. The density range explored in this simulation was from 0.07 to 0.17 and the graph indicates that the π distribution as determined from Eq. (5) did indeed cover the desired range in a uniform fashion.

IV. RESULTS AND DISCUSSION

We performed several TSMC simulations covering a substantial range of densities and temperatures during each run. During the course of each run we obtained free energies from Eq. (4) as well as canonical ensemble averages according to Eq. (7). We collected statistics for the region $0.035 < T^* < 1$ and $0.001 < \rho^* < 0.6$. To accelerate slow convergence at low temperatures, we also used multiparticle (or cluster) moves similar in spirit to the ones developed by Swendsen and Wang for lattice models,²² as in our previous work for three-dimensional ionic systems.²³ The energy summation was done by the Ewald method with vacuum boundary conditions, for the two-dimensional ion-ion interaction.²⁴ To estimate statistical uncertainties, we divided each run into five subruns and calculated the standard deviation of the subaverages.

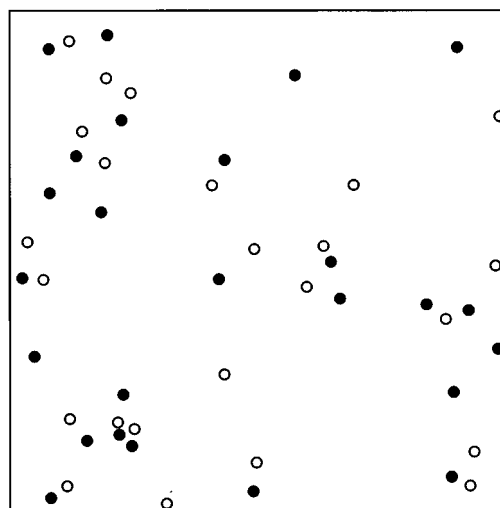


FIG. 2. Snapshot of an instantaneous system configuration at $T^*=1$, $\rho^*=0.025$, $N=50$ (conducting phase).

First we consider the low density metal-to-insulator transition. At high temperatures, the system consists of pairs and isolated ions. A typical snapshot of a conducting phase is shown in Fig. 2. At low temperatures, the strong attraction between positive and negative ions forces the system to associate. At very low densities one expects to see the system structure consisting of neutral pairs, but at slightly higher densities other neutral clusters will also be formed. A typical insulating configuration consisting of dimers and higher order clusters (tetramers, hexamers) is shown in Fig. 3. The concept of the insulator/conductor phase transition was first considered by Kosterlitz and Thouless.¹ This transition is generally thought to be of infinite order¹ and there are no observable singularities in the thermodynamic functions except a local maximum in the constant-volume heat

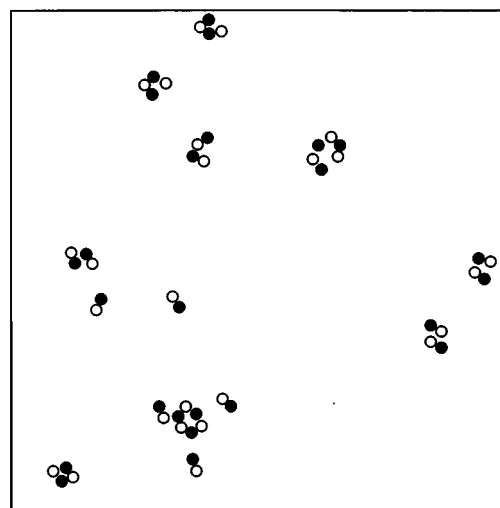


FIG. 3. Snapshot of an instantaneous system configuration at $T^*=0.065$, $\rho^*=0.025$, $N=50$ (dielectric phase).

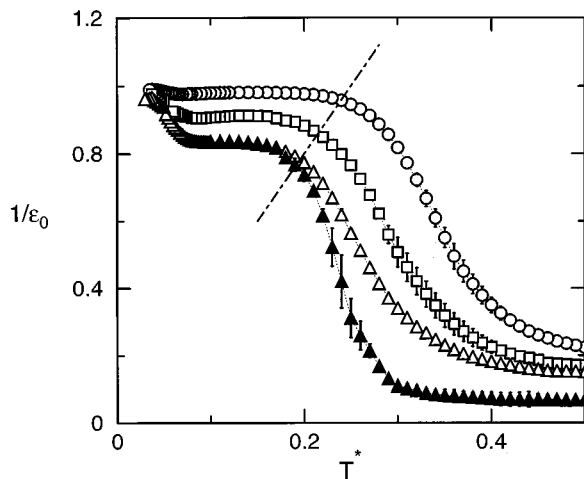


FIG. 4. Inverse dielectric function $1/\epsilon_0$ vs temperature for various densities and system sizes; ($\circ$) $\rho^*=0.001$, $N=50$; ($\square$) $\rho^*=0.005$, $N=50$; ($\triangle$) $\rho^*=0.01$, $N=50$; ($\blacktriangle$) $\rho^*=0.01$, $N=200$; dashed line: $1/\epsilon_0=4T^*$. The intersection of the curves with the line $4T^*$ gives an estimate of the KT transition temperature.

capacity.²⁵ According to the Kosterlitz–Thouless theory, the dielectric response of the system, which by standard linear response theory is given by

$$\frac{1}{\epsilon_0} = \lim_{k \rightarrow 0} \left[1 - \frac{2\pi}{k_B T} \frac{1}{k^2} \langle \hat{n}(\mathbf{k}) \hat{n}(-\mathbf{k}) \rangle \right], \quad (9)$$

jumps discontinuously from a finite value to zero as the transition temperature is approached from below. In Eq. (9), $\hat{n}(\mathbf{k})$ is the Fourier transform of the charge density, defined on the reciprocal lattice. When $\epsilon_0 = \infty$, this expression is the equivalent of the so-called Stillinger–Lovett sum rule.²⁶ According to Stillinger and Lovett,²⁶ an electrolyte always screens a small external electric field ($\epsilon_0 = \infty$). This is certainly the case for the conducting phase of the Coulomb gas, because of the abundance of the free ions. In contrast, the low density dielectric phase of this system, violates the Stillinger–Lovett sum rule due to the absence of isolated charge carriers. Nelson and Kosterlitz,³ using renormalization group arguments deduced that the size of the jump has a universal value

$$\frac{1}{\epsilon_0 T^*} = 4. \quad (10)$$

We directly evaluate $1/\epsilon_0$ using Eq. (9) for the first few allowed k -vectors in the periodic simulation cell.²⁷ Some characteristic curves are given in Fig. 4. Due to the finite number of particles used, we observe a continuous drop between two asymptotic values over a range of temperatures. Because of the roughening of the discontinuity, we estimate the transition points by the intersection of the $1/\epsilon_0$ curves with the line $4T^*$. At the limit of low density, $\epsilon_0 \approx 1$ for the insulating phase and from the universal criterion (10) it follows that $T^* \approx 1/4$.^{1–3} At higher densities, the concentration of bound pairs increases and the dielectric constant increases as well. This shifts the insulator–conductor transition to lower temperatures. Also, from Fig. 4 we see that at high temperatures

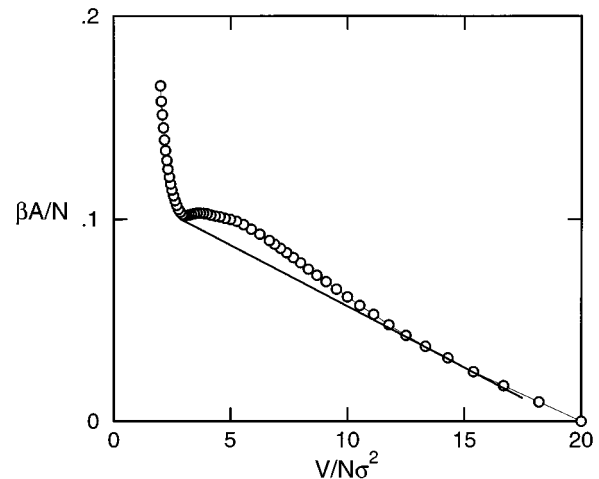


FIG. 5. Helmholtz free energy versus reduced molar volume for $T^*=0.05$ (open circles). Reference state at $V/N\sigma^2=20$. The tangent construction is indicated by the thick solid line.

the calculated $1/\epsilon_0$ curves asymptotically approach a density and system size dependent nonzero value. This is a consequence of imperfect screening for the finite simulated system. For larger systems (filled triangles in Fig. 4), the limiting value is smaller than for a small system at the same density (open triangles in Fig. 4). However, the KT transition temperature as seen in Fig. 4 is not too sensitive to the system size. We have performed simulations for $N=50$ and $N=200$ for all densities studied but do not show the data in Fig. 4 for clarity. The results for the KT transition can be seen in Fig. 6.

At finite densities and at low temperatures, our Monte Carlo data for the free energies show first-order phase separation between an insulating vapor phase and a conducting liquid. We estimate the coexisting densities by applying the double tangent criterion to the free energy data. This construction is shown in Fig. 5 for the $T^*=0.05$ isotherm. All calculations of the liquid–vapor phase envelope were performed for a $N=50$ system, since computing time was found to increase rapidly for higher system sizes. The calculated vapor–liquid coexistence curve is reproduced in Table I and Fig. 6, together with our predictions for the insulator–conductor line. We see that the first order phase transition occurs at low reduced temperatures and this situation is comparable to the range of vapor–liquid coexistence for the three-dimensional analog of the Coulomb gas, the “restricted primitive model.”^{14,23,28} In order to estimate the critical point, we fit the coexistence data to the classical scaling law for the density, $(\rho_1 - \rho_g) \propto (T_c - T)^{1/2}$, and the law of rectilinear diameters, $(\rho_1 + \rho_g)/2 \propto (T_c - T)$. Our estimate is, $T_c^* \sim 0.056$, $\rho_c^* \approx 0.21$. These values are only in rough agreement with that of Caillol and Levesque, $T_c^* \approx 0.087$ and $\rho_c^* \approx 0.14$, respectively.⁹ The quality of the coexistence data close to the critical point does not allow us to elucidate its character. The free energy curves tend to be flat and a tangent construction such as the one shown in Fig. 5 becomes ambiguous. Our near-critical coexistence data do not allow

TABLE I. Vapor–liquid coexisting properties for the two-dimensional Coulomb gas. T^* : coexistence temperature. ρ_g^*, ρ_l^* : density of saturated vapor and liquid, respectively. The numbers in subscripts indicate the uncertainty in the last digit.

T^*	ρ_g^*	ρ_l^*
0.035	0.002 ₁	0.55 ₁
0.036	0.005 ₁	0.54 ₁
0.037	0.006 ₁	0.53 ₁
0.038	0.0062 ₁₀	0.52 ₁
0.04	0.0075 ₁₀	0.48 ₄
0.0425	0.0125 ₁₀	0.47 ₅
0.045	0.030 ₅	0.45 ₁
0.0475	0.045 ₅	0.42 ₁
0.05	0.08 ₅	0.35 ₁
0.0525	0.105 ₁₀	0.33 ₂
0.055	0.14 ₂	0.30 ₃

us to distinguish between a normal coexistence curve and one with a cusp, such as the one obtained in Ref. 7. A cusp would indicate a tricritical point. In addition, size effects are expected to be pronounced in the neighborhood of the critical point. A formal finite-size analysis similar to that of Wilding and Bruce,²⁹ is under way and will be published in a future paper.

In summary, we have presented numerical results for the phase behavior of the two-dimensional Coulombic system. The calculated critical point for the vapor–liquid transition is located at $(T^*, \rho^*) = (0.056, 0.21)$, corresponding to significantly lower temperatures and higher densities than previous numerical and theoretical estimates for this model. The critical temperature is comparable and the critical density is higher than the corresponding three-dimensional system.^{14,23,28} The KT transition line extends from $T^* = 1/4$ at zero density to lower temperatures at higher densities and

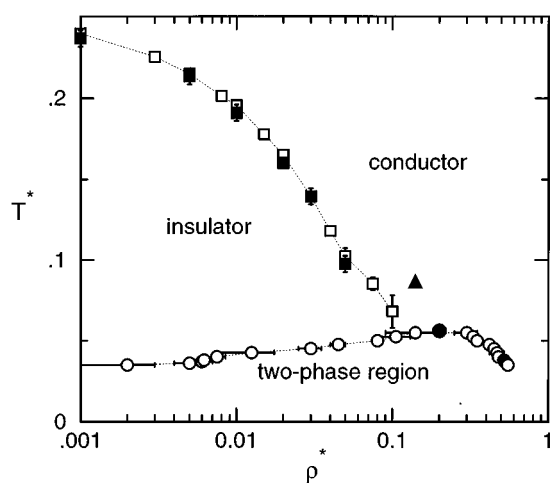


FIG. 6. Phase diagram of the two-dimensional Coulomb gas in the density-temperature plane. ($\square$) KT line, $N=50$; ($\blacksquare$) KT line, $N=200$; ($\circ$) first order vapor–liquid coexistence curve; ($\bullet$) estimate of the critical point from the present study; ($\blacktriangle$) critical point estimate of Caillol and Levesque (Ref. 9). Dotted lines are drawn for visual clarity.

transforms to first order vapor–liquid phase coexistence at finite densities.

The simulation method explored in this study, thermodynamic scaling Monte Carlo, appears to provide a direct path to the free energy of strongly interacting systems that have a tendency to form large-scale association structures. The method is likely to provide new insights into the behavior of other strongly interacting systems, such as dipolar systems, for which there are still significant uncertainties as to the existence of a well-defined vapor–liquid transition.²¹

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